AREC 570 Empirical Strategy Checklist

Unit 6 · Methods of Analysis

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Use this checklist as you are preparing to write your empirical section. This checklist follows the data checklist from Lecture 5.5. Rows marked **[6.x]** are covered in that lecture. Sources: **B** = Bellemare, "How to Write Applied Papers in Economics," Section 5; **AJAE** = AJAE Author Guidelines and Author's Final Checklist.

#	Relevant section	Purpose
1	Estimation strategy	**Prediction to equation [6.1].** Which testable prediction from your conceptual framework (Unit 4) each equation tests.
2	Estimation strategy	**Estimating equation [6.1, B].** The equation, with every variable and coefficient defined, subscripts ordered from smallest to largest unit, and a separate coefficient label for each specification.
3	Estimation strategy	**Unit of observation [6.1].** What one row is. It matches the observation unit and key in your data section (Unit 5).
4	Estimation strategy	**Parameter of interest [6.1].** Which coefficient answers the question, its units, how to read a one-unit change, and whether it has a causal interpretation (if not, what it describes).
5	Estimation strategy	**Estimator [6.1, B].** The estimation method (for example, OLS, linear probability model, probit, maximum likelihood, GMM) and why.
6	Estimation strategy	**Inference [6.1, B].** The null hypothesis; what the standard errors are robust to; the clustering level and why; how few clusters are handled; whether sampling weights are used.
7	Identification strategy	**Identifying assumption [6.1].** The assumption in one plain sentence, with no notation.
8	Identification strategy	**Identifying variation [6.1].** Which comparison (across units, over time, at a threshold, from assignment) moves the treatment and not the outcome directly.
9	Identification strategy	**Threats [6.1, B].** Whether each applies, and how the paper handles it in the design or a robustness check: reverse causality; unobserved heterogeneity; measurement error; spillovers across units or periods.
10	Depends on the design	**If the paper estimates a structural model [6.2].** The primitives; which variation identifies which parameter; the estimator in one paragraph, details in an appendix; fit or validation; the counterfactuals the model is used for.
11	Depends on the design	**If the paper uses a prediction step [6.3].** Which step is prediction and which is a causal parameter; how the predictor is tuned and evaluated; how the prediction error affects the parameter of interest.
12	Depends on the design	**If the paper collects primary data or runs an experiment [6.4, AJAE].** Assignment or sampling; sample size and why it is enough; IRB approval; the instrument or instructions in an appendix that does not identify the authors.
13	Depends on the design	**If the paper uses qualitative or mixed methods [6.4].** How cases or participants are selected; how evidence is coded; what question the qualitative evidence answers that the quantitative evidence cannot.
14	Scope and placement	**What the design does not identify [6.5].** Parameters, mechanisms, or populations outside the estimand; limits on external validity.
15	Scope and placement	**Robustness and sensitivity [6.5, B].** Three checks a referee would request, each tied to a threat in row 9 or to the inference choices in row 6.
16	Scope and placement	**Appendix [AJAE].** Which implementation details move to the appendix, cited in the text as an online appendix, not "available on request."
17	Scope and placement	**Replication [AJAE].** The data, code or pseudo-code, and README to be shared, or the confidentiality restriction to raise with the editor at submission.